

Segmentation

Deep Learning

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Lambda, HSE

Fast recap

Statistical learning theory

$\{(x_i, y_i)\}_{i=1}^{\ell}$ - training set $x \in \mathbb{R}^d$ - data features
 $y \in \mathbb{R}$ - target

$$\min_{\theta} \mathbb{E}_{(x,y) \sim \mathcal{D}} \mathcal{L}(\theta; x, y) \quad \mathcal{L}(\theta) = \frac{1}{n} \sum_{i=1}^n \ell(g_{\theta}(x_i), y_i)$$

STL

1. Problem is formulated in business language (e.g. “Highlight a tumor within MRI scan”)
2. Reformulate it formally in terms of math and STL (“given an image x , predict segmentation mask y ”)
3. Recall a universal approximation theorem
4. Remember that you have a prior knowledge about your data distribution (besides the normality assumption)
5. Fit your function f (your NN) to approximate target function g (your PDF or other law)
6. During fitting, you have to minimize the loss function (ERM). Once it minimized, and other formalities are satisfied, you are done!

Image segmentation

Computer Vision Tasks

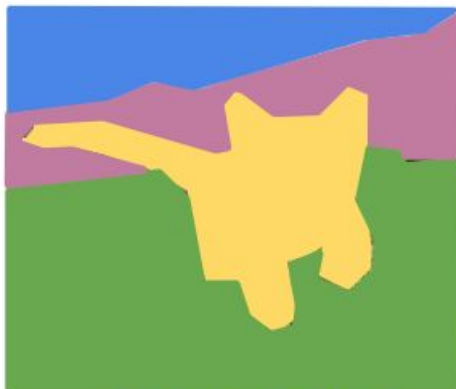
Classification



CAT

No spatial extent

Semantic Segmentation



GRASS, CAT,
TREE, SKY

No objects, just pixels

Object Detection



DOG, DOG, CAT

Multiple Object

Instance Segmentation



DOG, DOG, CAT

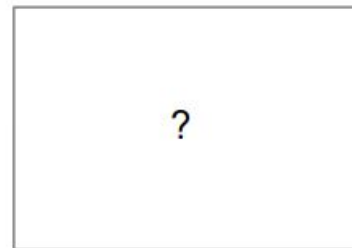
[This image is CC0 public domain](#)

Formulation



GRASS, CAT,
TREE, SKY, ...

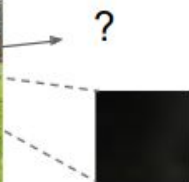
Paired training data: for each training image,
each pixel is labeled with a semantic category.



At test time, classify each pixel of a new image.

Problem

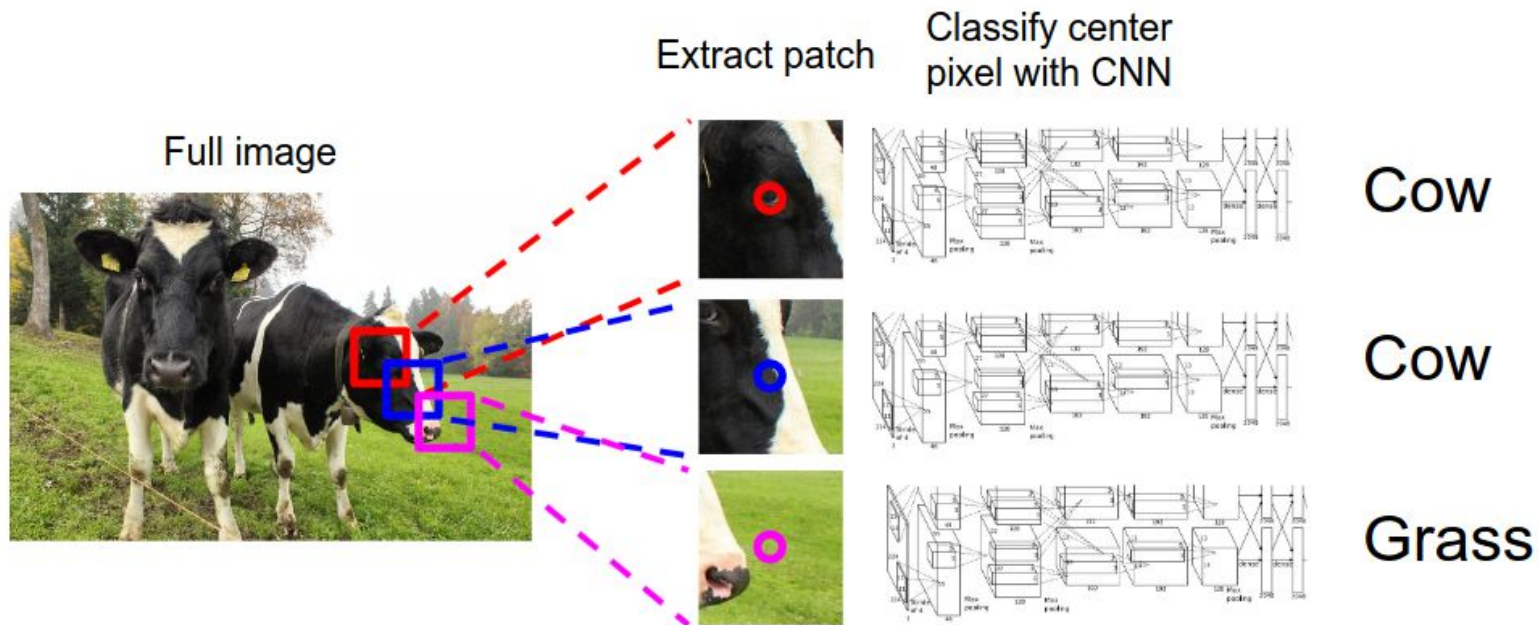
Full image



Impossible to classify without context

Q: how do we include context?

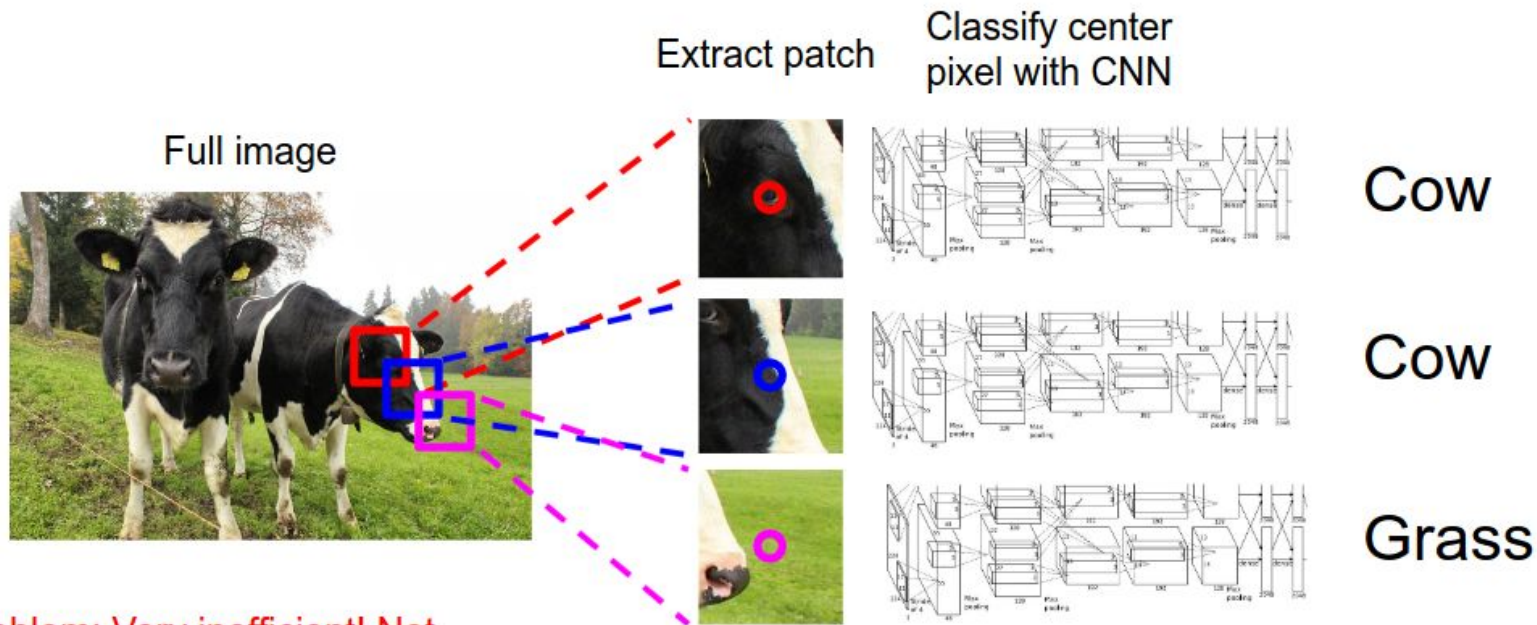
Context



Farabet et al, "Learning Hierarchical Features for Scene Labeling," TPAMI 2013

Pinheiro and Collobert, "Recurrent Convolutional Neural Networks for Scene Labeling", ICML 2014

Inefficiency

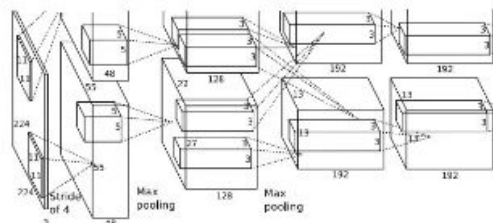


Problem: Very inefficient! Not reusing shared features between overlapping patches

Farabet et al, "Learning Hierarchical Features for Scene Labeling," TPAMI 2013
Pinheiro and Collobert, "Recurrent Convolutional Neural Networks for Scene Labeling", ICML 2014

Shared Convolutional features

Full image

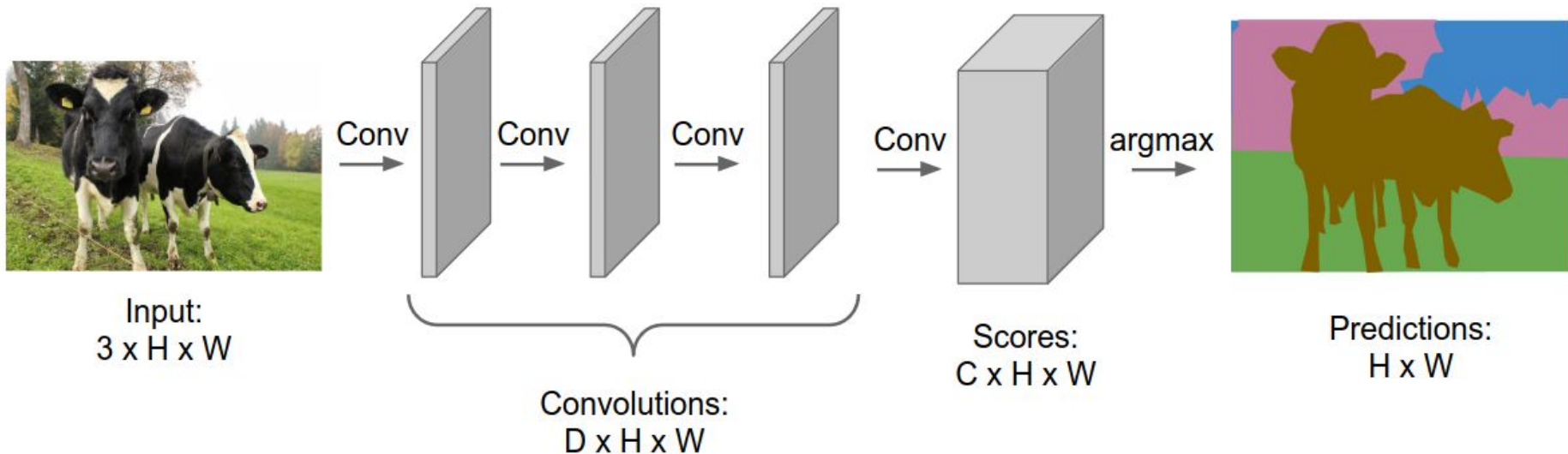


An intuitive idea: encode the entire image with conv net, and do semantic segmentation on top.

Problem: classification architectures often reduce feature spatial sizes to go deeper, but semantic segmentation requires the output size to be the same as input size.

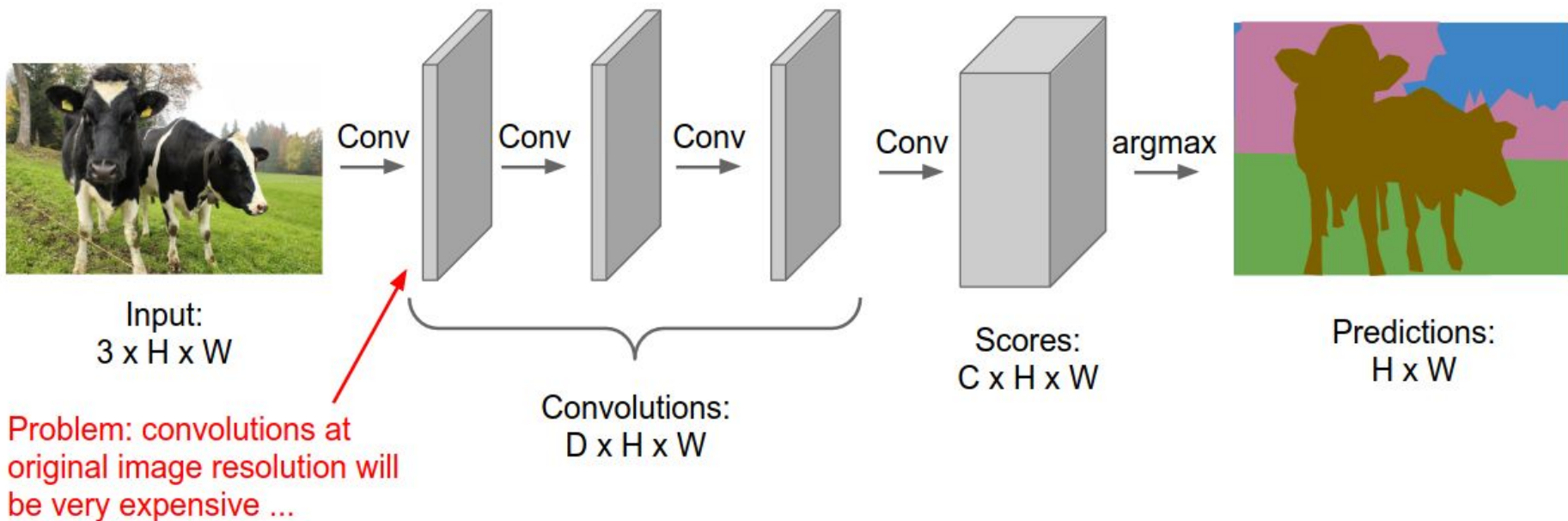
CNN Segmentation

Design a network with only convolutional layers without downsampling operators to make predictions for pixels all at once!



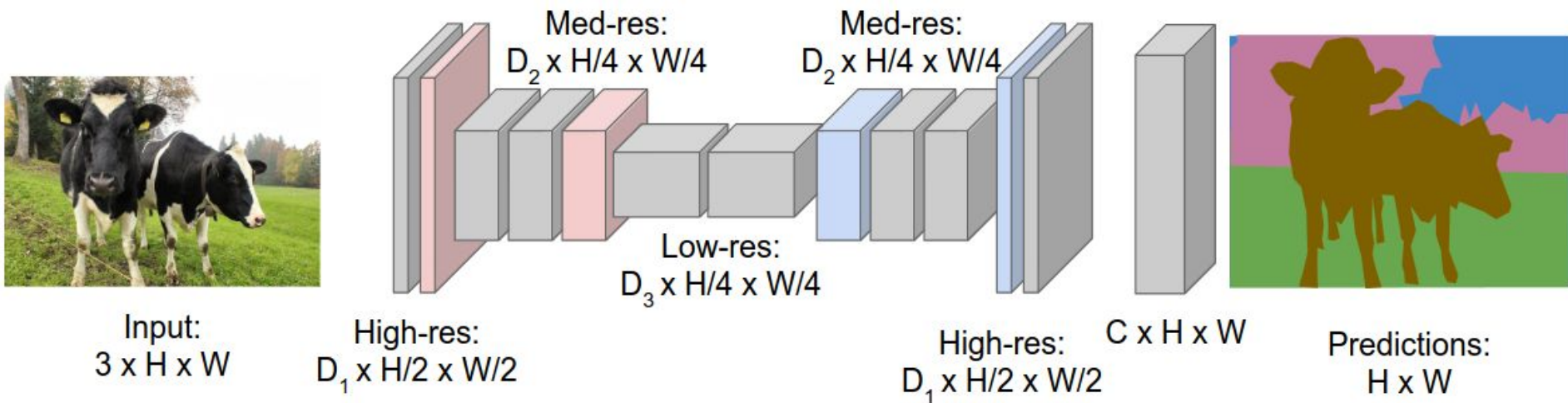
CNN Segmentation

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CNN Segmentation

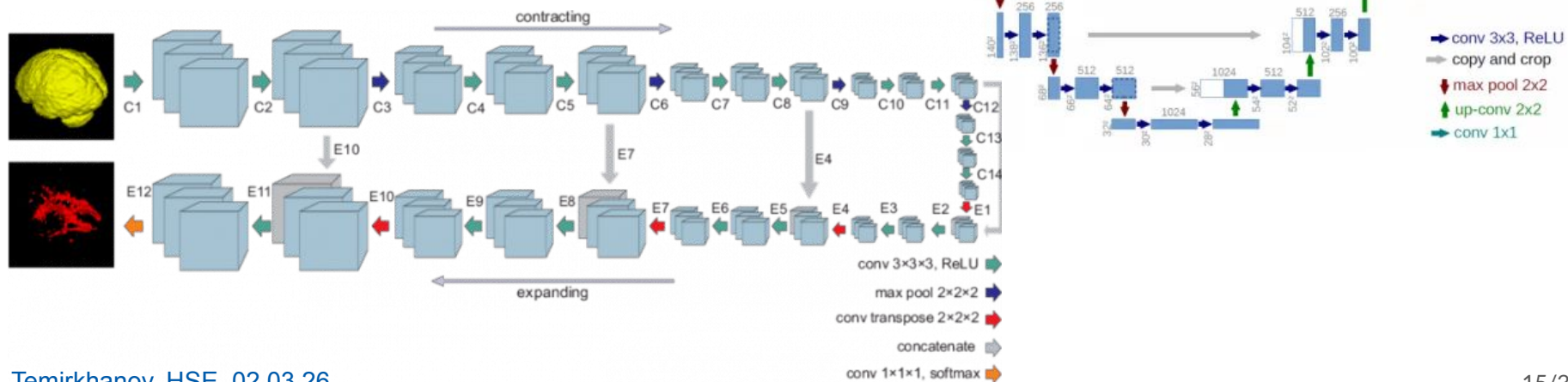
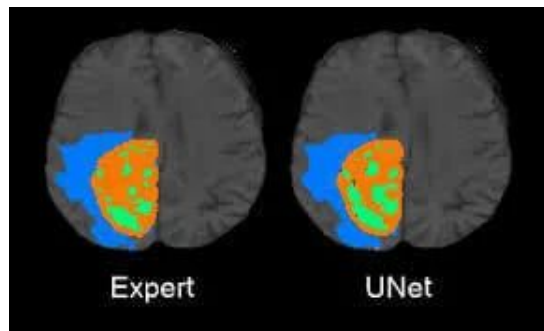
Design network as a bunch of convolutional layers, with **downsampling** and **upsampling** inside the network!



Long, Shelhamer, and Darrell, "Fully Convolutional Networks for Semantic Segmentation", CVPR 2015

Noh et al, "Learning Deconvolution Network for Semantic Segmentation", ICCV 2015

UNet



Unpooling

Nearest Neighbor

1	2
3	4

Input: 2 x 2



1	1	2	2
1	1	2	2
3	3	4	4
3	3	4	4

Output: 4 x 4

"Bed of Nails"

1	2
3	4

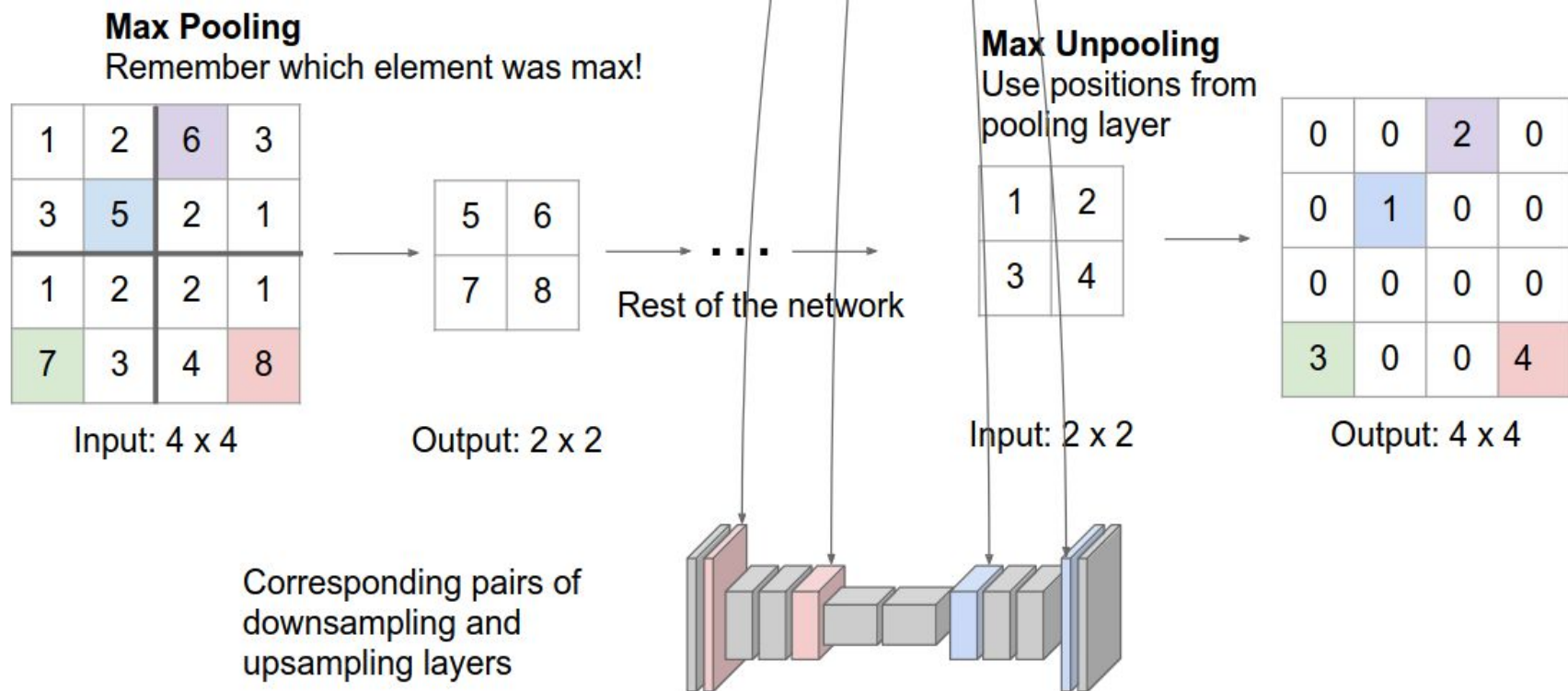
Input: 2 x 2



1	0	2	0
0	0	0	0
3	0	4	0
0	0	0	0

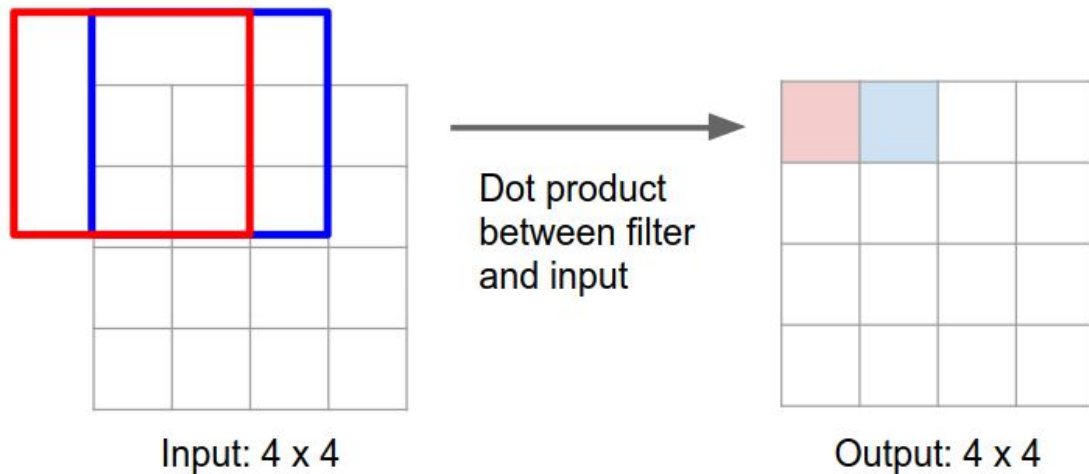
Output: 4 x 4

Max unpooling



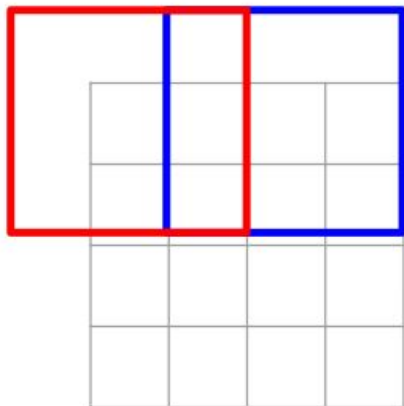
Transpose Convolution

Recall: Normal 3 x 3 convolution, stride 1 pad 1



Transpose Convolution

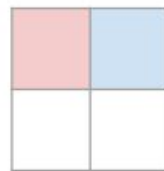
Recall: Normal 3 x 3 convolution, stride 2 pad 1



Input: 4 x 4



Dot product
between filter
and input



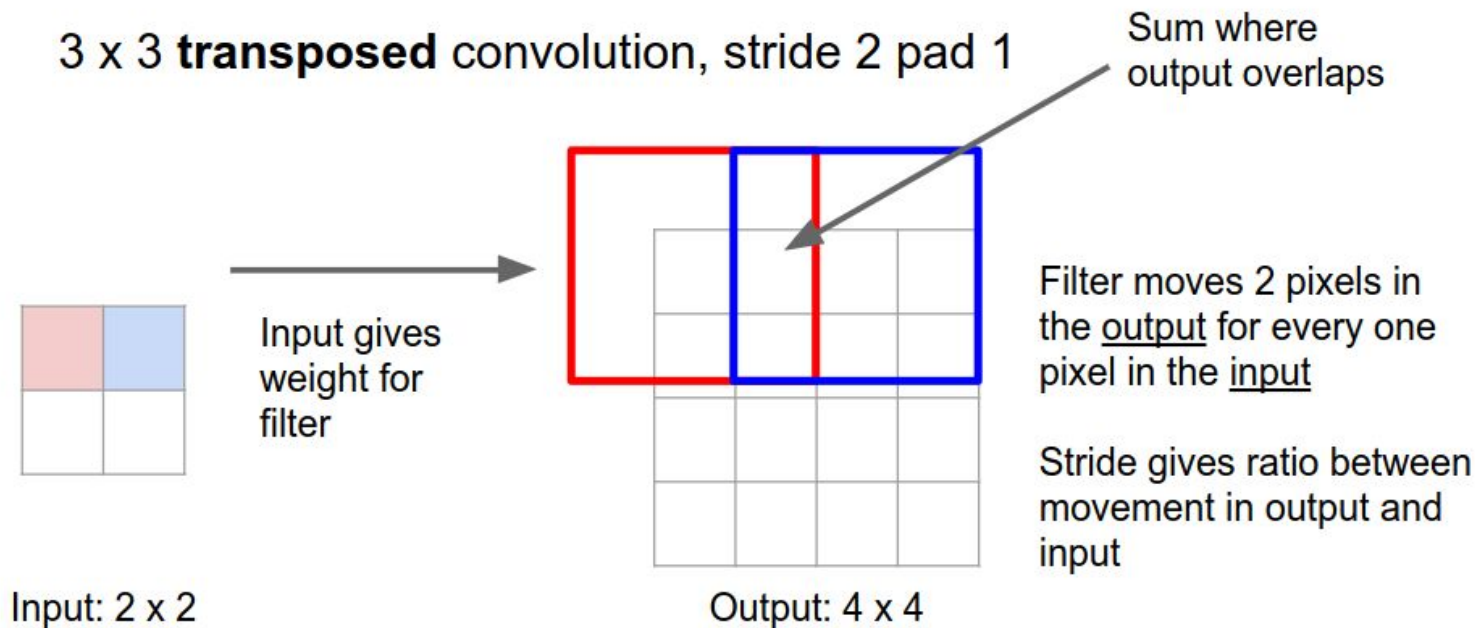
Output: 2 x 2

Filter moves 2 pixels in
the input for every one
pixel in the output

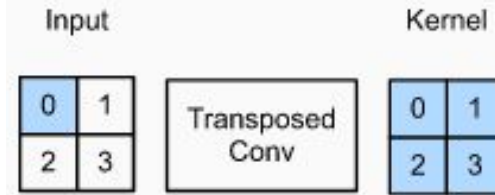
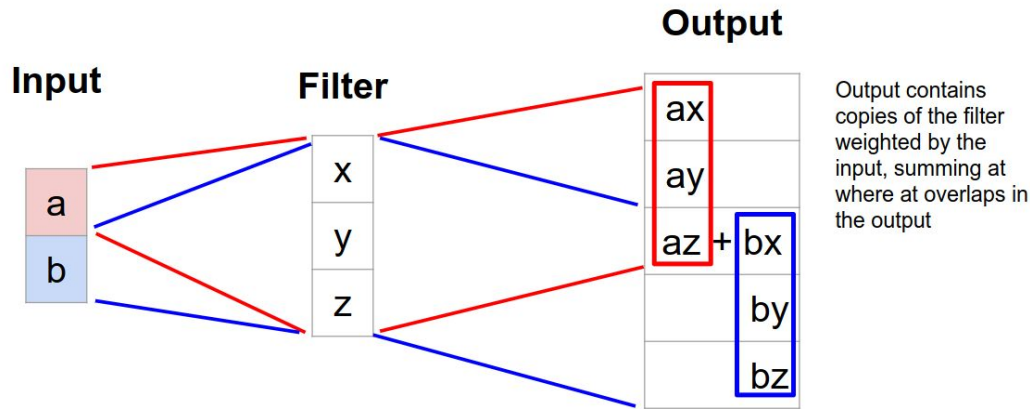
Stride gives ratio between
movement in input and
output

We can interpret strided
convolution as “learnable
downsampling”.

Transposed Convolution



Transposed Convolution



Output

$$\begin{bmatrix} 0 & 0 & \\ 0 & 0 & \\ & & \end{bmatrix} + \begin{bmatrix} & 0 & 1 \\ & 2 & 3 \\ & & \end{bmatrix} + \begin{bmatrix} & & \\ 0 & 2 & \\ 4 & 6 & \end{bmatrix} + \begin{bmatrix} & & \\ & 0 & 3 \\ & 6 & 9 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 4 & 6 \\ 4 & 12 & 9 \end{bmatrix}$$

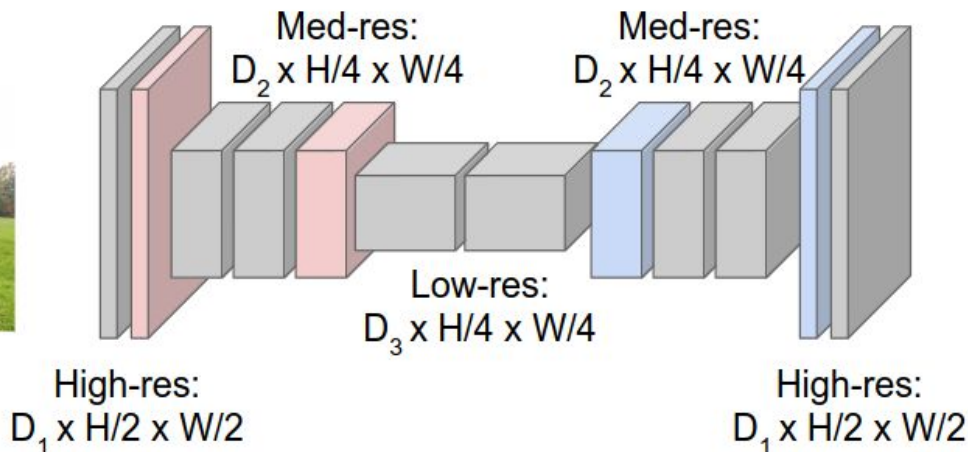
Semantic Segmentation

Downsampling:
Pooling, strided
convolution



Input:
 $3 \times H \times W$

Design network as a bunch of convolutional layers, with **downsampling** and **upsampling** inside the network!



Upsampling:
Unpooling or strided
transposed convolution



Predictions:
 $H \times W$

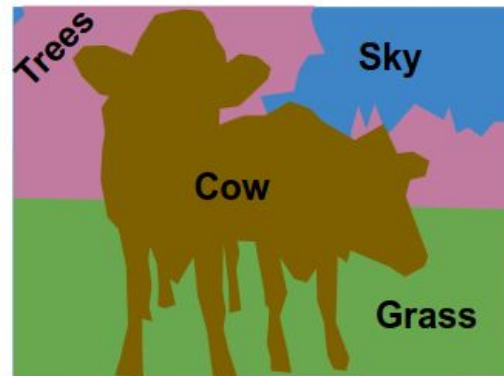
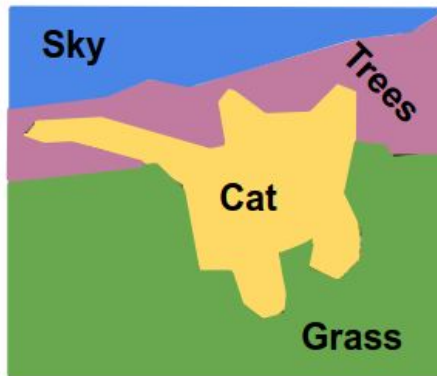
Long, Shelhamer, and Darrell, "Fully Convolutional Networks for Semantic Segmentation", CVPR 2015

Noh et al, "Learning Deconvolution Network for Semantic Segmentation", ICCV 2015

Semantic Segmentation

Label each pixel in the image with a category label

Don't differentiate instances, only care about pixels



Instance Segmentation

Classification



CAT

No spatial extent

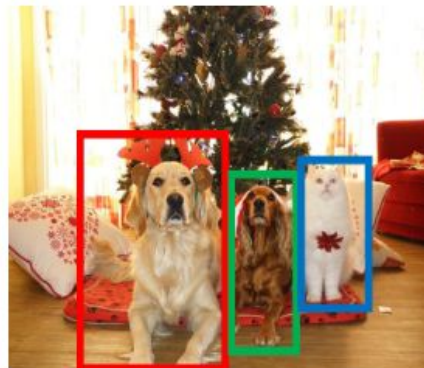
Semantic Segmentation



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TREE, SKY

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Object Detection



DOG, DOG, CAT

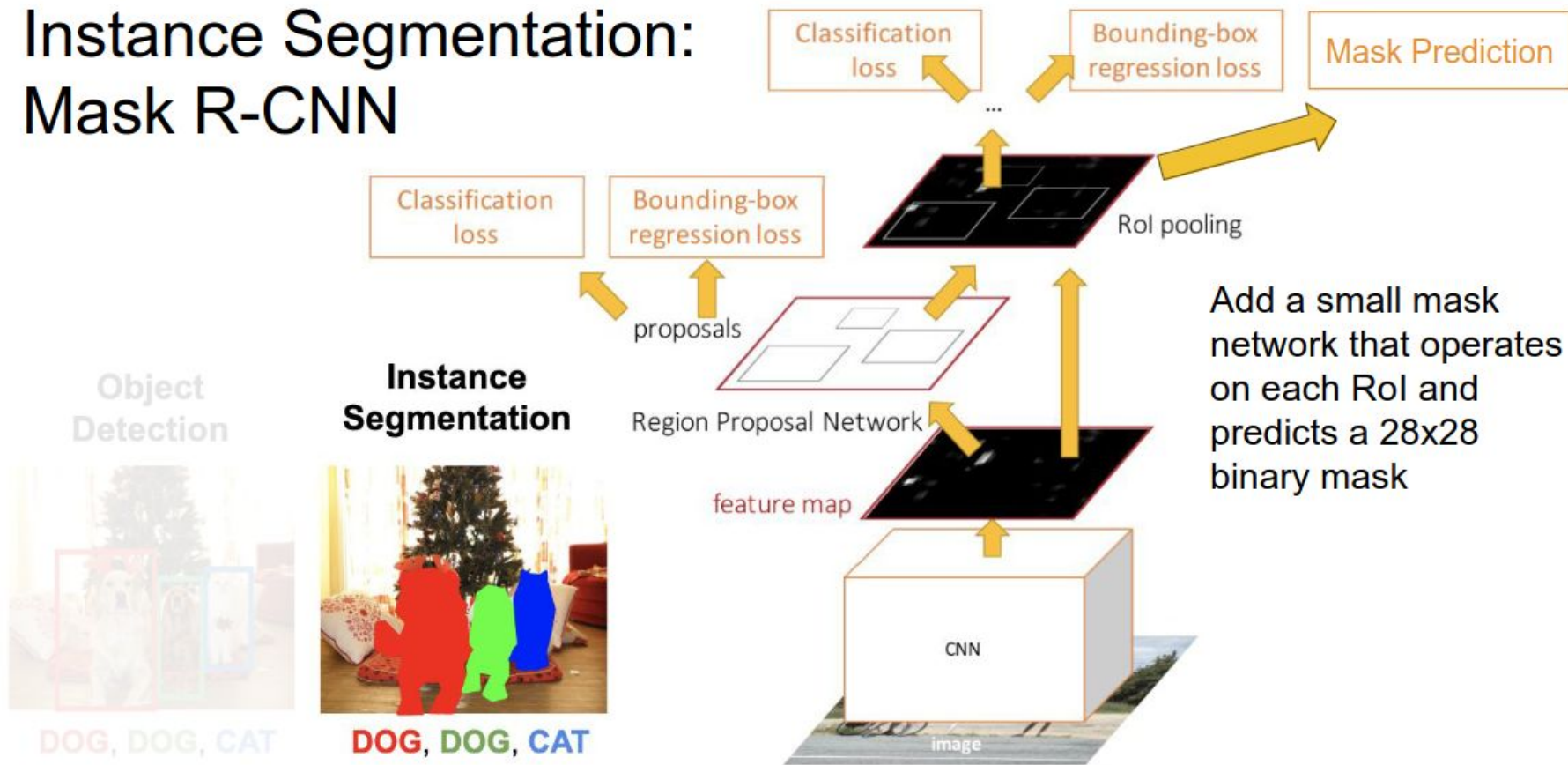
Multiple Object

Instance Segmentation



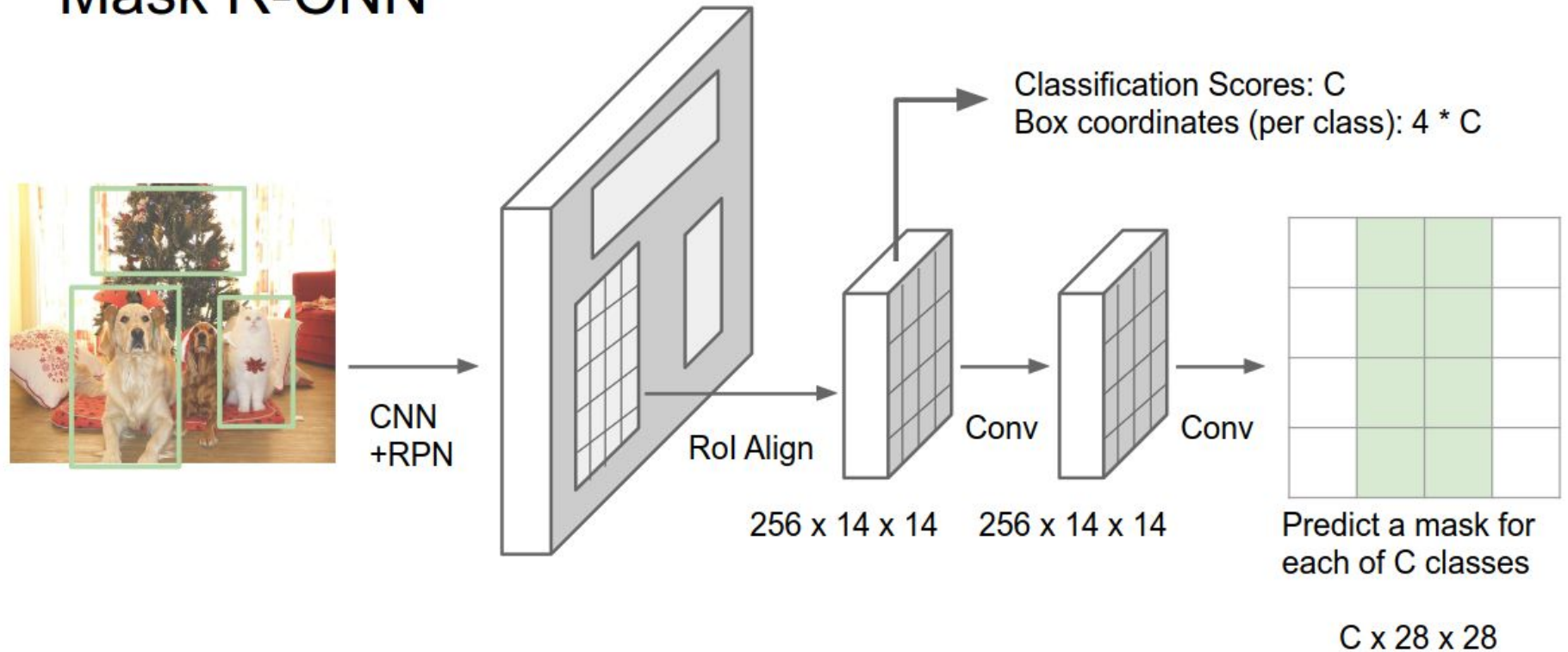
DOG, DOG, CAT

Instance Segmentation: Mask R-CNN



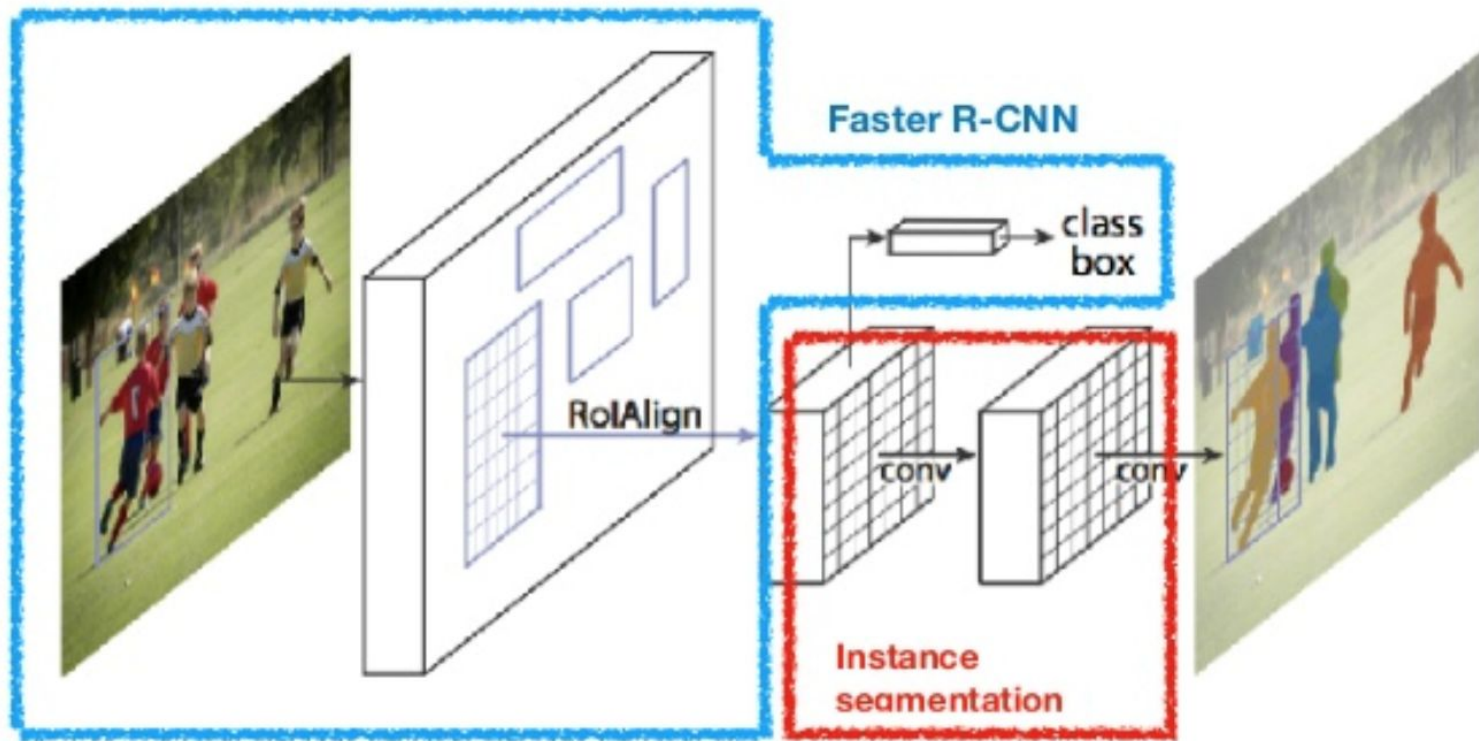
He et al, "Mask R-CNN", ICCV 2017

Mask R-CNN

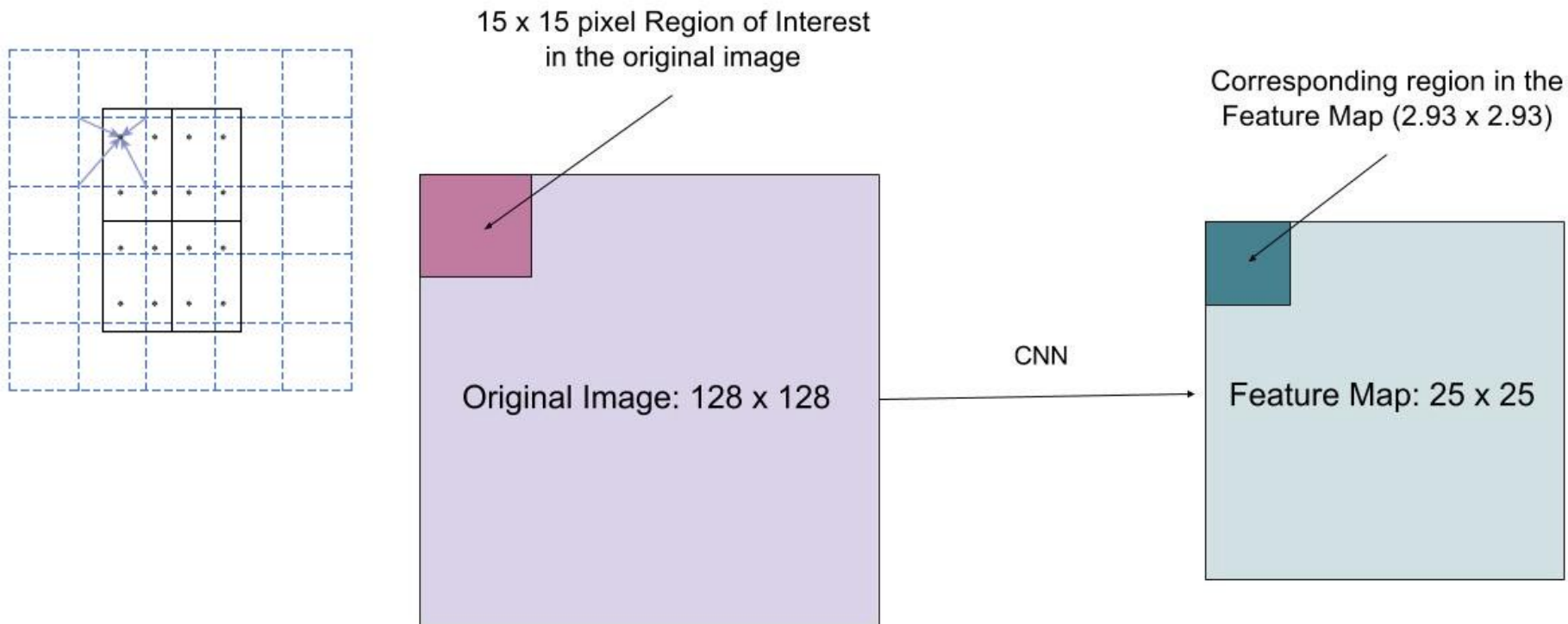


He et al, "Mask R-CNN", arXiv 2017

Mask R-CNN



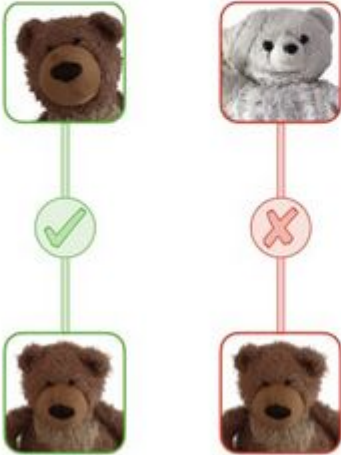
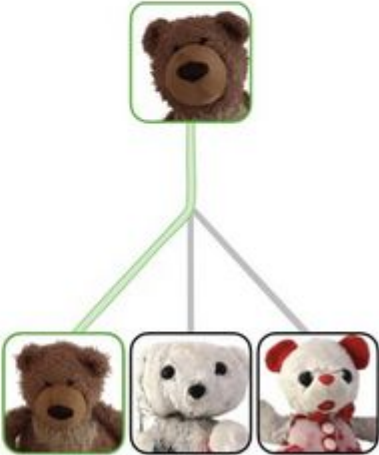
ROI Alignment



Bonus part

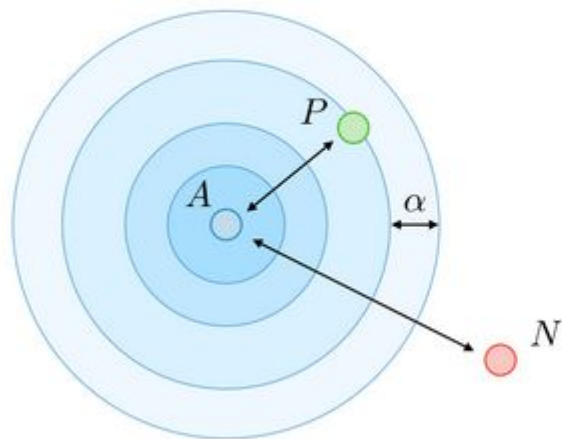
Addition vision tasks

Face verification and recognition

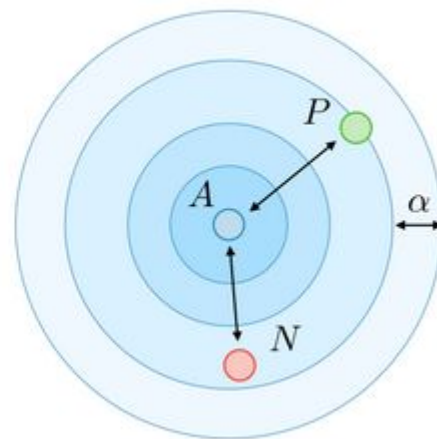
Face verification	Face recognition
• Is this the correct person? • One-to-one lookup	• Is this one of the K persons in the database? • One-to-many lookup
Query  Reference The diagram illustrates the face verification process. It shows two vertical paths. The left path starts with a 'Query' image of a brown teddy bear, followed by a green checkmark icon, and then a 'Reference' image of the same brown teddy bear. The right path starts with a 'Query' image of a grey teddy bear, followed by a red 'X' icon, and then a 'Reference' image of a brown teddy bear. This represents a one-to-one comparison.	Query  Database The diagram illustrates the face recognition process. It shows a 'Query' image of a brown teddy bear at the top. Below it, three lines branch out to a 'Database' containing three images: a brown teddy bear, a white teddy bear, and a white teddy bear with red ears. This represents a one-to-many search.

Triplet Loss

$$\ell(A, P, N) = \max(d(A, P) - d(A, N) + \alpha, 0)$$



$$\ell(A, P, N) = 0$$



$$\ell(A, P, N) > 0$$

Style Transfer and Domain Adaptation



Content C

+



Style S

=



Generated image G